

Summary

ADAS and Embedded Applications Engineer (**B.Tech, IIT Madras**) with 3 years of specialized R&D experience at Bajaj Auto. Expert in designing and deploying ADAS/EV control systems, sensor fusion (Radar/Camera/IMU), and real-time C/C++/Python applications for embedded platforms, like Arm, Renesas, and Infineon. Praised for **AI-assisted automation** of classical workflows and process improvement.

Education and Scholastic Achievements

Program	Institution	CGPA	Year of completion
B. Tech (Hons) in Mechanical Engineering	Indian Institute of Technology Madras	8.92	2023
• Was awarded a minor in Systems Engineering for completing 4 graduate-level courses in estimation, controls, optimization, and ML			

Skills

- **Programming & Frameworks** : C, C++, Python, MATLAB, Simulink, Stateflow, Shell Scripting
- **Communication Protocols** : CAN (High-speed/FD), IPC (Inter-Process Communication), UART, SPI, I2C, LIN
- **Embedded Systems & Hardware** : TI mmWave Radar, ARM Cortex-M, Renesas RH850, Raspberry Pi, ESP32, STM32
- **Control & Estimation** : IMM-EKF, Sensor Fusion, PID, MPC, Kalman Filtering
- **ADAS & Perception** : clustering, multi-object tracking, real-time object detection, and classification
- **Engineering Tools** : Code Composer Studio, TargetLink, Embedded Coder, Git, Linux, vSignalizer

Professional Experience**Bajaj Auto Technology Ltd (Research and Development) (08/2023 - Present)****ADAS algorithm design and testing: (06/2025 - Present)**

- **Developed** the company's first **end-to-end ADAS perception stack** on the **TI AWR1843** platform, and established tools for CI/CD workflows
- **Architected** **real-time sensor fusion** on the onboard **DSP**, synchronizing **IMU data**, and **vehicle odometry** from the vehicle **CAN bus** with **radar point clouds** for clustering (**DBSCAN**) and tracking (**JPDA**). Designed an **Interacting Mixture Model (IMM)** comprising **Extended Kalman Filters (EKF)** for **Constant Acceleration (CA)** and **Constant Turn (CT)** models for track maintenance and classification
- **Optimized** the perception and tracking **algorithms** to run on constrained hardware with zero heap allocation and in under **120kB of stack**
- Developed a **Hardware-In-Loop (HIL)** setup to simulate road data from vehicle logs on the MCU, cutting down time spent on validation
- Implemented active safety features such as **Active Emergency Braking** and **Adaptive Cruise Control** using track collision probability
- Developed Python-based software utilities using AI coding assistants, replacing commercial tools and **reducing licensing costs by 50LPA**

EV Control Strategy: (08/2023 - 05/2025)

- Deployed a **sensor-less mass and grade estimation** algorithm for a shared mobility project using existing CAN bus data, enabling the company to remotely detect and prevent vehicle overloading of field vehicles without the need for additional hardware or manpower.
- **Engineered** a custom **Coordinate Transformation library** in MATLAB to normalize raw IMU readings from the sensor frame to the vehicle frame across multiple SKUs. Integrated a self-correcting algorithm to mitigate variance in mounting solutions. Used the processed IMU data for **firmware-level IMU applications** (roll-over detection, blinker auto-cancellation, pitch-based regen/hill-hold) to the vehicle's production software, adding advanced dynamics features while achieving a 5% reduction in electronics BOM by eliminating redundant sensors
- **Resolved 20+ critical firmware defects** in traction control and hill-assist logic through rigorous unit testing and system-level debugging

Engine Testing: (Internship) (05/2022 - 07/2022)

- Built the **software infrastructure** for organizing and analyzing historic engine oil reports using **statistical methods**, and created an interface for visualizing cylinder block surface roughness data as cumulative frequency distribution curves, before and after the stress test.
- Was awarded a **pre-placement offer** for introducing new methods of analysis and streamlining existing workflows across departments

AI and Engineering Projects**Final Year Project (VECH Lab, IIT Madras) (10/2022 - 05/2023)****Control of camber morphing wing**

- Spearheaded a research project in collaboration with **DRDO** to develop the **control system** for a **composite camber morphing airfoil**
- Architected a real-time control loop for a camber-morphing airfoil, modelled as a finite-state unsteady airfoil combining full-state feedback with a Kalman Filter to drive a Model Predictive Control (MPC) with a prediction horizon of 5 steps and a control horizon of 2 steps.
- Set up a **Hardware in Loop (HIL)** across MATLAB, Arduino, motor driver, and actuator for validating and improving the control logic

Personal Projects: Edge AI Applications**Road-Anomaly Detection & Geospatial Mapping**

- Developed an Edge AI road-anomaly detection system using **STM32 X-CUBE-AI** to deploy an optimized **Neural Network** on an STM32 microcontroller; integrated 6-axis **IMU (MPU6050)** data for high-frequency vibration analysis and synchronized **GPS (NEO-6M)** coordinates to automate the geospatial mapping of road hazards like pot-holes and speed-bumps.